

SECURITY TOOL REPORT

Password Strength Analyzer & Social Engineering Mitigation

Project Overview

As part of the cybersecurity initiatives at **CyberSecure Consulting Group**, this tool was developed to mitigate risks associated with weak authentication. The script performs real-time static analysis of user credentials against industry-standard entropy requirements.

Core Security Features

Length Verification	Enforces 8+ character minimum to prevent rapid brute-force.
Complexity Audit	Checks for Alpha (Upper/Lower), Numeric, and Special Character diversity.
Common Password Filter	Cross-references inputs against a blacklist of top 20 breached passwords.

Technical Implementation (AST Logic)

The script utilizes regular expressions and boolean logic gates to provide granular feedback. Below is a snippet of the heuristic evaluation logic:

```
def check_numbers_special(password):  
    has_number = any(char.isdigit() for char in password)  
    has_special = bool(re.search(r"[!@#$%^&*()]", password))  
    return (has_number and has_special)
```

Risk Recommendations

Based on the analysis, users are provided with specific remediation steps:

- **Entropy Improvement:** Use a mix of character sets to increase keyspace complexity.
- **Blacklist Avoidance:** Do not use dictionary-based words or common sequences like '123456'.
- **Standard Alignment:** Aim for passwords that trigger the "Strong Password" result.